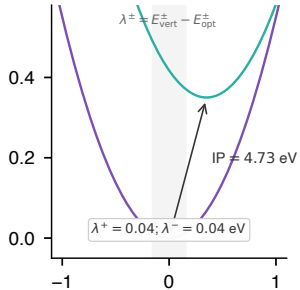


charged GEO_OPT + vertical SP at neutral geometry); compare dopant-dependent IP/EA and $\lambda^\pm$.

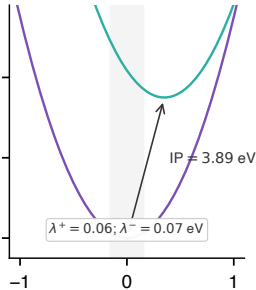
energy (eV, schematic)

Pristine

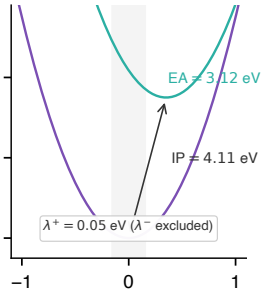
$$\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$$



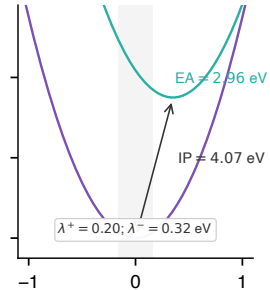
N



B



P



configuration coordinate Q